<u>Annexure-II</u>

Salient features of the varieties recommended for release/notification by the 18th meeting of Central Sub-Committee on Crop Standards, Notification and Release of Varieties for Horticulture Crops held under the chairmanship of Dr. H. P. Singh, ICAR on 20th January, 2012 at IIVR, Varanasi.

S.No.	Particulars	Salient features
1. (i)	Crop	Garlic *(Allium sativum)*
ii)	Variety	VL Lahsun 2 (VGP 5)
iii)	Institution or Agency responsible for developing the variety.	Vivekananda Parvatiya Krishi Anusandhan Sansthan (ICAR), Almora-263601, (Uttarakhand)
iv)	Parentage with details of its pedigree.	Selection from garlic germplasm available in the institute.
v)	Agency responsible for maintaining breeder seed.	Vivekananda Parvatiya Krishi Anusandhan Sansthan (ICAR), Almora-263601, (Uttarakhand)
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None, neither in plant type nor bulb.
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in the States of Jammu & Kashmir, Himachal Pradesh and Uttarakhand under both irrigated and rainfed conditions.
viii)	Description of the variety/hybrid. a) Plant height (cm) b) Distinguishing morphological characteristics. Description of variety 1. Plant growth habit 2. Av. Plant height (cm) 3. Leaf attitude 4. Leaf shape 5. Leaf cross-section 6. Pseudo stem colour 7. Pseudo stem length (cm) 8. Neck thickness (cm) 9. Leaf colour 10. Leaf waxy ness 11. Av. Leaf length (cm) 12. Av. Leaf width (cm) 13. Av. No. of leaves/plant 14. Bulbils position 15. Ability to produce scape 16. Ability to flower 17. Bulb shape	 30-60 • Plants are semi erect, vigorous in growth with green foliage. • Bulb creamy-white violet stripes. Semi erect 30-60 (it may vary as per the altitude) Semi erect Partly curved V-shaped Green with purple trips 5 – 9.0 1.5 – 2.0 Green Present 45 – 55 (Fourth leaf) 2 – 3 (Fourth leaf) 8 – 9 Top Yes No Oblate

	18. Bulb type 19. Bulb colour 20. Bulb size 21. Polar dia(mm) 22. Equitorial dia(mm) 23. Av. Bulb weight/plant (g) 24. Clove skin colour 25. Clove type 26. Clove skin thickness 27. Av. Clove length (cm) 28. Av. Clove width (cm) 29. Clove flesh colour 30. Av. No. of cloves/bulb Maturity (range in number of days) seeding/transplanting to flowering/fruiting. Maturity on ground (early/medium/late- wherever classification exists). Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.). Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality. c) Reaction to environmental stress.	Compact Creamy – White with purple trip Big 36 – 45 (It may vary as per the altitude) 38 – 60 (It may vary as per the altitude) 40 – 70 (It may vary as per the altitude) Creamy – White Big Thin 2.1 – 4.0 1 – 1.9 Creamy – White 11 – 15 This variety is responsive to nitrogenous fertilizer and a dose of 100 kg N, 50 kg P and 50 kg K. It posses higher bulb weight (40- 70 g/bulbs) and High TSS (37.3 Brix)
	Description of the parents/variety	Not applicable
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	Resistant to purple blotch (Alternaria porri) and stemphylium blight disease.
x)	Reaction to major pests (under field and controlled conditions including stored pests)	--
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	150-240 qtl/ha.

S.No.	Particulars	Salient features
2. (i)	Crop	Bottle Gourd Hybrid *{Lagenaria siceraria (Mol) Standl}*
ii)	Variety	Santosh 20 (KBGH 20)
iii)	Institution or Agency responsible for developing the variety.	Krishidhan Vegetable Seeds India Pvt. Ltd. 5th floor, Sai Capital, Opp. ICC Complex, Senapati Bapat Road, Shivaji Nagar, Pune – 411016 (Maharashtra)
iv)	Parentage with details of its pedigree.	KVBG 53 x KVBG 11
v)	Agency responsible for maintaining breeder seed.	Krishidhan Vegetable Seeds India Pvt. Ltd., At. Vadhu Khurd, Post Fulgaon, Tah. Haveli Dist. Pune - 412316
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	Pusa Naveen
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in the States of Delhi, Uttar Pradesh, Bihar and Jharkhand under rainy and summer season.
viii)	Description of the variety/hybrid. a) Plant height (m) b) Distinguishing morphological characteristics.	 4.5 - 5.0 i. Cylindrical fruits ii. Medium long fruit (34 – 45 cm) iii. Green fruit colour iv. Blossom end fruit shape: Semi-blunt
	Description of parents a) Leaf colour b) 50% flowering c) Days to first picking d) Fruit length e) Fruit diameter f) Fruit shape g) Hairiness of fruit h) Fruit colour i) Average fruit wt.	 Green 53 – 55 days 60 – 62 days 16" – 18" long 3" – 4" Cylindrical Hairy fruit Green 500 – 550 g
	Maturity (range in number of days) seeding/transplanting to flowering/fruiting.	
	Maturity on ground (early/medium/late- wherever classification exists).	Days to first picking: 55 – 60 days
	Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.).	Good yield at 25 tonnes of FYM plus 80:120:80 kg NPK/ha, spacing 6 feet x 2 feet, June sowing for rainy season crop and February sowing for summer season and winter season.

	Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality. c) Reaction to environmental stress.	Fruits have good keeping quality.	
	Description of the parents/variety a) Leaf colour b) 50% flowering c) Days to first picking d) Fruit length e) Fruit diameter f) Fruit shape g) Hairiness of fruit h) Fruit colour i) Average fruit wt.	**Female** Green 55 – 57 days 63 – 65 days 18" – 20" long 3" – 4" Bottle shape fruit notch at the neck of fruit. Medium hairy Green 500 – 600 g	**Male** Light green 53 – 55 days 60 – 62 days 15" – 18" long 4" – 5" Cylindrical bulging at neck. Hairy fruit Light green 400 – 500 g
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	Tolerant to powdery and downy mildew.	
x)	Reaction to major pests (under field and controlled conditions including stored pests)	No major incidence of pests seen.	
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	350 – 400 q/ha.	

S.No.	Particulars	Salient features
3. (i)	Crop	Coconut *(Cocos nucifera L.)*
ii)	Variety	Kalpa Samrudhi (IND 376)
iii)	Institution or Agency responsible for developing the variety.	Central Plantation Crops Research Institute, Kasaragod and Horticultural Research Station, AAU, Kahikuchi, Guwahati, Assam.
iv)	Parentage with details of its pedigree.	Selection from IND 058 (Malayan Yellow Dwarf) was used as female parent and selection from IND 069 (West Coast Tall) was used as male parent.
v)	Agency responsible for maintaining breeder seed.	Central plantation Crops Research Institute, Kasaragod.
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in coconut growing tracts of West Coast region and Assam.
viii)	Description of the variety/hybrid. a) Plant height	 The palms are semi tall and attain an average height

		of 733.82 cm 28 years after planting.
	b) Distinguishing morphological characteristics.	The palms are semi tall without prominent bole. The colour of the petiole is green. The variety bears green coloured, oval fruits and the dehusked fruits are found in shape. The seedlings are vigourous, producing more number of leaves within 12 months, having more leaf area and dry weight.
	c) Characteristics different from those of the parents.	The female parent palms are dwarf, takes 66 months for flowering, bears bright yellow fruits and has yellow petiole colour. The male parent palms are tall, bearing green fruits and takes about 98 months for flowering. The palms of IND 376 bear bright green fruits; have bright green petioles; semi tall habit, takes 45 months for initiation of flowering; and profuse bearing.
	Description of variety	
	1. Age of measurement (years)	28
	2. Category	Semi Tall
	3. Crown shape	Circular
	4. Presence of bole	Absent
	5. Plant height (cm)	733.82
	6. Girth of trunk (cm)	69.36
	7. Total number of leaves	33
	8. Length of petiole (cm)	122.27
	9. Length of leaflet bearing portion (cm)	376.45
	10. Number of leaflets	109
	11. Length of leaflet (cm)	122.73
	12. Breadth of leaflet (cm)	6.01
	13. No. of leaf scars in 1 m length	25
	14. Length of 10 internodes (cm)	39.18
	15. Age at 50% flowering (months)	60
	16. Length of inflorescence (cm)	88.64
	17. Length of spikelet bearing portion (cm)	40.36
	18. Length of stalk (m)	48.27
	19. Length of spikelet (cm)	41.09
	20. No. of spikelet's in the inflorescence	37.18
	21. Number of female flowers	26
	22. Number of inflorescences on the crown	14
	23. Colour and shape of fruit	Green, Oval
	24. Length of fruit (cm)	19.38
	25. Breadth of fruit (cm)	15.44
	26. Weight of fruit (g)	1032.33
	27. Thickness of husk (cm)	
	Apical	6.00
	Distal	2.75
	Equatorial	2.35
	28. Shape of nut	Round

	29. Weight of nut (g) 30. Percent of husk to whole fruit weight 31. Thickness of kernel (cm) 32. Weight of kernel (g) 33. Weight of copra (g) Maturity (range in number of days) seeding/transplanting to flowering/fruiting. Maturity on ground (early/medium/late- wherever classification exists). Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.). Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality. Reaction to environmental stress.	681.33 34.0 1.77 359.67 219.46 The palms are regular bearers and commence flowering in 45 months after planting under rain fed conditions. The average time taken for flowering of 50% of the palms in the hybrid population is 60 months. Not applicable Higher nitrogen use efficiency was observed in the seedling of the proposed hybrid. The average quantity of tender nut water is 346 ml. Based on the organoleptic test; the tender nut water is classified as "good" in taste with a TSS of 6^0 Brix. The tender nut water has Na content of 35.1 ppm and K content of 2370 ppm. The oil extracted from the copra of this hybrid is having 45.4 % lauric acid. Tolerant to drought
	Description of the parents/variety	--
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	No major disease out break observed under field conditions.
x)	Reaction to major pests (under field and controlled conditions including stored pests)	No Major pest attacks observed under field conditions.
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	117 nut/palm/year and copra 4.5 t/ha.

S.No.	Particulars	Salient features
4. (i)	Crop	Coconut *(Cocos nucifera L.)*
ii)	Variety	Kalpa Sankara (CGD x WCT hybrid)
iii)	Institution or Agency responsible for developing the variety.	Central Plantation Crops Research Institute, Regional Station, kayamkulam, Alappuzha District, Kerala.
iv)	Parentage with details of its pedigree.	Hybrids produced by crossing between palms of Chowghat Green Dwarf and West Coast Tall which

		are high yielding and free from root (wilt) disease.
v)	Agency responsible for maintaining breeder seed.	Central Plantation Crops Research Institute, Regional Station, kayamkulam, Alappuzha District, Kerala.
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in root (wilt) disease prevalent areas of the country.
viii)	Description of the variety/hybrid. a) Plant height	 The palm attains a height of around 4.98 m at 18 years of age.
	b) Distinguishing morphological characteristics.	--
	Description of variety	
	1. Age of measurement (years)	18 Years
	2. Category	Semi Tall
	3. Crown shape	Circular
	4. Presence of bole	Yes
	5. Plant height (cm)	498
	6. Girth of trunk (cm)	78.7
	7. Total number of leaves	30.3
	8. Length of petiole (cm)	115.3
	9. Length of leaflet bearing portion (cm)	375
	10. Number of leaflets	115.2
	11. Length of leaflets (cm)	113.8
	12. Breadth of leaflet (cm)	5.2
	13. No. of leaf scars in 1 m length	20.1
	14. Length of 10 internodes (cm)	46.6
	15. Age at firs flowering (months)	36-40
	16. Length of inflorescence (cm)	112.0
	17. Length of spikelet bearing portion (cm)	59.1
	18. Length of stalk (m)	62.9
	19. Length of spikelet (cm)	39.5
	20. No. of spikelet in the inflorescence	38.7
	21. Number of female flowers	23.2
	22. Number of inflorescences/year	15.4
	23. No. of bunches/year	15.4
	24. No. of nuts/year	84
	25. Bunches with button	4.6
	26. Bunches with nuts	10.8
	27. Quantity of water (ml)	373
	28. Sweetness of the water (7-8 months old)	3 (Sweet)
	29. Sweetness of the meat	3 (Sweet)
	30. Colour of fruit	Greenish Brown
	31. Shape of fruit	Oval
	32. Length of fruit (cm)	18.7
	33. Breadth of fruit (cm)	14.3

	34. Weight of fruit (cm) 35. Thickness of husk (cm) 36. Weight of nut (gm) 37. Percent of husk to whole fruit weight 38. Thickness of kernel (cm) 39. Weight of kernel (gm) 40. Thickness of shell (cm) 41. Copra content (g/nut) 42. Copra/palm/year (kg) 43. Copra/ha (t) 44. Oil content (%) 45. Oil/ha (t) Maturity (range in number of days) seeding/transplanting to flowering/fruiting. Maturity on ground (early/medium/late- wherever classification exists). Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.). Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality. Reaction to environmental stress.	839.2 4.4 527.8 37.5 1.18 295.7 0.27 170 14.62 2.500 67.5 1.690 Comes to flowering by 36-40 months after planting. Early • High yield (14,700 nuts/ha/year) and early hearing (36-40 months after planting). • Tolerance to root (wilt) disease. The quantity of tender nut water is 373 ml and sweet in taste. Tolerance to moisture stress gives better yield under rainfed conditions in farmer's plots in the root (wilt) disease prevalent tract.
	Description of the parents/variety	Chowghat Green Dwarf as female parent and West Coast tall as male parent. Parental palms were selected from hotspots of root (wilt) disease and the parental palms were disease free and high yielding.
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	--
x)	Reaction to major pests (under field and controlled conditions including stored pests)	Requires adequate plant protection measures against major pests particularly red palm weevil when large scale plantings are adopted.
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	84 nuts/palm/year and cobra 2500 kg/ha.

S.No.	Particulars	Salient features
5. (i)	Crop	Coconut *(Cocos nucifera L.)*
ii)	Variety	Kalpasree (Chowghat Green Dwarf)
iii)	Institution or Agency responsible for developing the variety.	Central Plantation Crops Research Institute, Regional Station, kayamkulam, Alappuzha District, Kerala.
iv)	Parentage with details of its pedigree.	Selection from the Chowghat Green Dwarf variety grown in farmers plots.
v)	Agency responsible for maintaining breeder seed.	Central Plantation Crops Research Institute, Regional Station, kayamkulam, Alappuzha District, Kerala.
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in root (wilt) diseased tracts of the country.
viii)	Description of the variety/hybrid. a) Plant height	The palm attains a height of around 4 m at 20 years of age.
	b) Distinguishing morphological characteristics.	--
	Description of variety	
	1. Age of measurement (years)	20 Years
	2. Category	Dwarf
	3. Crown shape	Circular
	4. Presence of bole	No
	5. Plant height (cm)	400
	6. Girth of trunk (cm)	59.92
	7. Total number of leaves	25.28
	8. Length of petiole (cm)	110.49
	9. Length of leaflet bearing portion (cm)	315.03
	10. Number of leaflets	105.2
	11. Length of leaflets (cm)	101.85
	12. Breadth of leaflet (cm)	4.29
	13. No. of leaf scars in 1 m length	40.22
	14. Length of 10 internodes (cm)	25.96
	15. Pollination group	Self Pollinated
	16. Age at firs flowering (months)	40
	17. Length of inflorescence (cm)	59.00
	18. Length of spikelet bearing portion (cm)	27.30
	19. Length of stalk (m)	31.70
	20. Length of spikelet (cm)	29.60
	21. No. of spikelet in the inflorescence	26.30
	22. Number of female flowers	17.62
	23. Number of inflorescences/year	9.10
	24. No. of bunches/year	9.10
	25. No. of nuts/year	90
	26. Bunches with button	4.3

	27. Bunches with nuts 28. Quantity of water (ml) 29. Sweetness of the water (7-8 months old) 30. Sweetness of the meat 31. Colour of fruit 32. Shape of fruit 33. Length of fruit (cm) 34. Breadth of fruit (cm) 35. Weight of fruit (cm) 36. Thickness of husk (cm) 37. Weight of nut (gm) 38. Percent of husk to whole fruit weight 39. Thickness of kernel (cm) 40. Weight of kernel (gm) 41. Thickness of shell (cm) 42. Copra content (g/nut) 43. Copra/palm/year (kg) 44. Copra/ha (t) 45. Oil content (%) 46. Oil/ha (t)	4.8 172.3 4 (Very Sweet) 4 (Very Sweet) Green Oval 17.0 12.0 683.4 4.2 349.5 47.8 1.02 171.2 0.23 96.3 8.667 2.045 66.5 1.360
	Maturity (range in number of days) seeding/transplanting to flowering/fruiting.	Comes to flowering by 36-40 months after planting.
	Maturity on ground (early/medium/late- wherever classification exists).	Early
	Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.).	Low incidence of root (wilt) disease has been reported early flowering (36-40 months after planting)
	Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality.	The quantity of tender nut water is 240 ml and very sweet in taste. The nutritive value of tender nut water; Total sugars-4.80 g/ml, Potassium content-2150 ppm, Sodium content- 22.40 ppm, TSS – 4.8 Brix. Data on Fatty acid profile reveals that Kalpasree is rich is Long chain Unsaturated Fatty Acids (LUSFA's) and is healthier. Kalpasree oil is having – 25% to 40% more essential fatty acids during all seasons compared to oil from WCT and COD, respectively. Besides, Kalpasree oil is rich in essential fatty acids especially Linoleic acid.
	Reaction to environmental stress.	Kalpasree is having better recovery potential in terms of lipid peroxidation, photosynthesis and Water Use Efficiency (WUE) WUE by about 40%.
	Description of the parents/variety	Not applicable
ix)	Reaction to major diseases under field	--

	and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	
x)	Reaction to major pests (under field and controlled conditions including stored pests)	Caution is advised to adopt plant protection measures against major pests particularly red palm weevil, when large scale commercial plantings are adopted.
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	76 nuts/palm/year and copra 2045 kg/ha & 1.360 tonnes oil/ha.

S.No.	Particulars	Salient features
6. (i)	Crop	Chilli *(Capsicum annuum L.)*
ii)	Variety	Kashi Gaurav (VR-338)
iii)	Institution or Agency responsible for developing the variety.	Indian Institute of Vegetable Research P. B. No. 01, P.O.- Jakhini (Shahanshahpur) Varanasi – 221 305 (U.P.), India.
iv)	Parentage with details of its pedigree.	Selection from local collection.
v)	Agency responsible for maintaining breeder seed.	Indian Institute of Vegetable Research P. B. No. 01, P.O.- Jakhini (Shahanshahpur) Varanasi – 221 305 (U.P.), India.
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in the States of West Bengal and Assam in autumn and winter season.
viii)	Description of the variety/hybrid. a) Plant height (cm) b) Distinguishing morphological characteristics.	 50-60 The plants are dushy, tolerant to thrips and mites, dark green foliage, very good general combiner, 50% flowering in 35-40 days after transplanting, fruits and dark green and dark red when ripe, 9-11 cm long and 1.1-1.2 cm thick, pedant and pungent.
	Maturity (range in number of days) seeding/transplanting to flowering/fruiting.	Red ripe maturity after 95-100 days of seeding and 65-70 days after transplanting.
	Maturity on ground (early/medium/late- wherever classification exists).	Medium maturity
	Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.).	Nursery sowing: mid July to mid August Planting distance – 60 x 45 cm Fertilizer – 100:60:60 NPK kg/ha Seed rate – 500 g per hectare
	Quality attributes purpose for which recommended: Table/processing/export/any other.	Suitable for table purpose

	Table quality attributes/Nutritional quality. Reaction to environmental stress.	 Sturdy plants can withstand low temperature (87.23% pollen viability at 10^0 C)
	Description of the parents/variety	Not applicable
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	Tolerant to anthracnose
x)	Reaction to major pests (under field and controlled conditions including stored pests)	Less attack of thrips
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	95 – 99 qtl/ha.

S.No.	Particulars	Salient features
7. (i)	Crop	Chilli *(Capsicum annuum L.)*
ii)	Variety	Kashi Sinduri (IVPBC-535)
iii)	Institution or Agency responsible for developing the variety.	Indian Institute of Vegetable Research P. B. No. 01, P.O.- Jakhini (Shahanshahpur) Varanasi – 221 305 (U.P.), India.
iv)	Parentage with details of its pedigree.	Introduction from AVRDC, Taiwan & Selection.
v)	Agency responsible for maintaining breeder seed.	Indian Institute of Vegetable Research P. B. No. 01, P.O.- Jakhini (Shahanshahpur) Varanasi – 221 305 (U.P.), India.
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in the States of Karnataka, Tamil Nadu and Kerala in autumn-winter Season.
viii)	Description of the variety/hybrid. a) Plant growth habit b) Distinguishing morphological characteristics. c) Characteristics different from those of the parents. Maturity (range in number of days) seeding/transplanting to flowering/fruiting. Maturity on ground	 Determinate The Plants are determinate, spreading type, resistant to anthracnose disease, green foliage, behaves as maintainer of cytoplasmic male sterile lines, 50% flowering in about 30-35 days after transplanting, fruits and green at fresh stage and dark red at ripe stage, 10-11 cm long and 1.1-1.4 cm thich, pendant behavior of fruiting. Original material was more pungent. Red ripe maturity after 95-100 days of seeding and 65-70 days after transplanting. Medium maturity

	(early/medium/late- wherever classification exists). Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.). Quality attributes purpose for which recommended: Table/processing/export/any other. Table quality attributes/Nutritional quality. Reaction to environmental stress.	 Nursery sowing: mid July to mid August Planting distance – 60 x 45 cm Fertilizer – 100:60:60 NPK kg/ha Seed rate – 500 g per hectare Suitable for oleoresin extraction due to high colour value and less pungency. 88.57% Pollen viability at 10^0 C.
	Description of the parents/variety	Not applicable
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	Tolerant to Anthracnose
x)	Reaction to major pests (under field and controlled conditions including stored pests)	Less attack of thrips
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	131 – 155 qtl/ha.

S.No.	Particulars	Salient features
8. (i)	Crop	Sponge gourd *(Luffa cylindrical L.)*
ii)	Variety	Kashi Divya (VR-I)
iii)	Institution or Agency responsible for developing the variety.	Indian Institute of Vegetable Research, PB – 01, PO – Jakhini (Shahanshahpur), Varanasi-221305 (U. P.) India
iv)	Parentage with details of its pedigree.	VRSG-101 x VRSG-111
v)	Agency responsible for maintaining breeder seed.	Indian Institute of Vegetable Research, PB – 01, PO – Jakhini (Shahanshahpur), Varanasi-221305 (U. P.) India
vi)	State the varieties, which most closely resemble with the proposed variety in general, characteristics.	None
vii)	Specific area of adaptation and recommended ecology.	Recommended for release and cultivation in the States of Punjab, Uttar Pradesh, Bihar and Jharkhand in rainy and summer season.
viii)	Description of the variety/hybrid. a) Plant growth habit b) Distinguishing morphological characteristics. i. Stem colour ii. Plant type iii. Fruit shape iv. Fruit colour	 Viny, vine length 3.5 – 5.25 m Green Medium (3.5-5.25 m) Cylindrical Light green

	v. Fruit size	--	
	vi. Node number at which 1st female flower appears	8 – 9	
	vii. No. of fruits per plant	14 – 16	
	viii. Number of picking	5 – 6 times	
	ix. Days to first harvest	48 – 50 days	
	x. Fruit length	20.0 – 25.0 cm	
	xi. Fruit diameter	4.36 cm	
	xii. Fruit weight	125.36 g	
	xiii. Fruit pulp colour	White	
	xiv. Days taken for physiological maturity (Seed to seed)	120 – 125 days	
	Maturity (range in number of days) seeding/transplanting to flowering/fruiting.	Seeding to fruit maturity 115-120 days	
	Maturity on ground (early/medium/late- wherever classification exists).	Medium maturity	
	Agronomic features (e.g. resistance to lodging, shattering, fertilizer responsiveness suitability for early or late sown conditions seed rate etc.).	• It can be grown in summer and rainy season. • It can be grown as mixed or inter crop. • Suitable for river bed cultivation • Suitable for trellies system • Responsiveness to higher does of fertilizer of 80:60:60 kg N.P.K./ha • Seed rate: 3.5 kg ha^{-1} • Sowing time – First fort night of July and First fort night of February. • Spacing – 3.5 m row to row and 75 cm plant to plant.	
	Quality attributes purpose for which recommended: Table/processing/export/any other.	Vegetable purpose and distance marketing (export).	
	Table quality attributes/Nutritional quality.	Good cooking quality and very tasty	
	Reaction to environmental stress.	More temperature tolerant and late sowing (September – October and March – April)	
	Description of the parents/variety	**Female**	**Male**
	a) Number of branches/vine	8 – 10	9 – 12
	b) Foliage colour	Light green	Dark green
	c) Stem colour	Light green	Dark green
	d) Average fruit weight (g)	130	165
	e) Fruit length (cm)	25	28
	f) Fruit diameter (cm)	9.00	10.00
	g) Fruit colour	Light green	Light green
	h) Number of fruits/plant	12	8
	i) Yield (q/ha)	219.00	224.00
	j) Days to 50% harvest	65 – 70	75 – 80

	k) Maturity/ripening (range in number of days seed ton be seed) l) Is there any problem of synchronization? If yes, methods of over it. m) Reaction to major diseases (under field and controlled conditions-reaction to physiological strains/races, biotypes to be indicated wherever possible) n) Reaction to major pests (under field and controlled condition) o) Agronomic features (e.g. fertilizer responsiveness, seed rate etc) p) Reaction to environmental stress	130 – 135 No Susceptible to Powdery mildew and Downy mildew Less Leaf minor Seed rate 3.5 kg/ha More temp. tolerant	140 – 145 No • No Powdery mildew • Less downy mildew Susceptible to leaf minor Seed rate 3.5 kg/ha More temp. tolerant
ix)	Reaction to major diseases under field and controlled conditions (reaction to physiological strains/races/bio-types to be indicated wherever possible.	Resistance to Anthracnose disease and tolerant to downy and powdery mildew.	
x)	Reaction to major pests (under field and controlled conditions including stored pests)	Tolerant to leaf minor, fruit fly and Red pumpkin beetle.	
xi)	Average yield under normal condition (yield in kg/qtl./ton per ha.)	260 qtl/ha.	